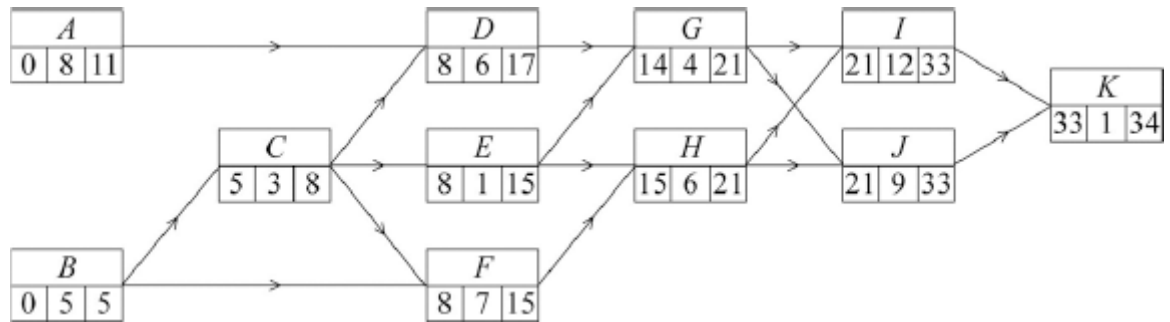


Mark schemes

Q1.

(a)



Forward pass, correct at two of D, E, F

All correct

Backward pass, correct at G **AND** H ft

All correct

M1

A1

M1

A1

4

(b)

Activity	Predecessor
A	—
B	—
C	B
D	A, C
E	C
F	B, C
G	D, E
H	E, F
I	G, H
J	G, H
K	I, J

6+ correct

B1

All correct

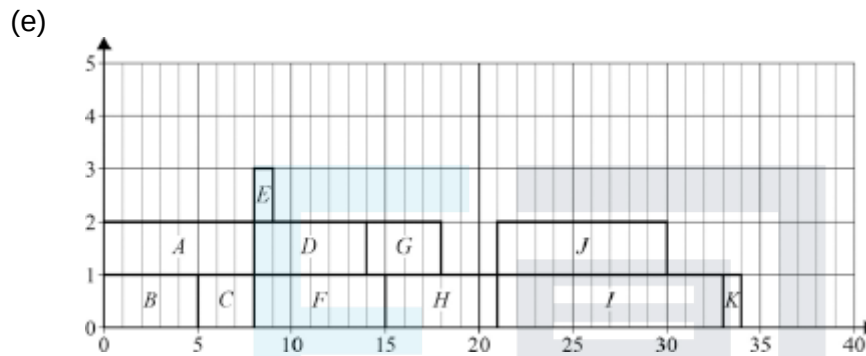
B1
2

(c) (Critical) $B C F H I K$

B1
1

(d) (Float E) 6 (hrs)

B1
1



Their critical activities and 3 others shown

M1

Critical activities and 3 others correct

A1

All correct, condone floats seen

A1
3

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(f) 34 (hrs)

B1
1

(g) 62 (hrs)

B1
1

[13]

Q2.

(a) $x = 4$

B1

$y = 12$

B1

$z = 13$

B1

3

- (b) $B D H J$ and $C E I J$
first correct path

M1

2nd correct and no others

A1

2

- (c) G

B1

Float = 3

B1

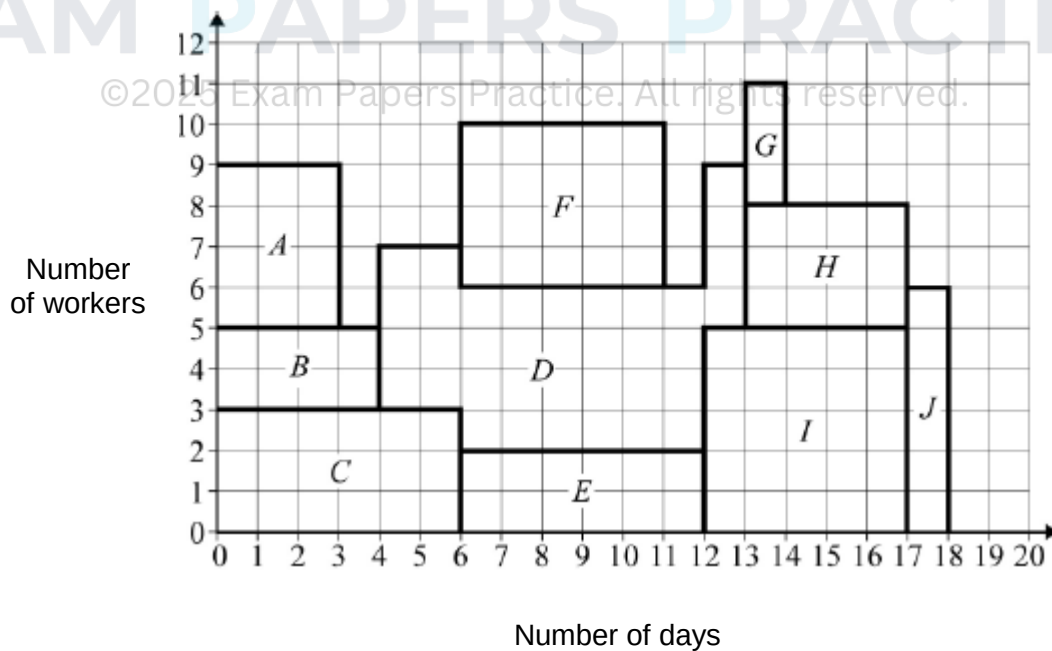
2

- (d) One of their CPs correct height
and correct durations

M1

B, D, H, J and C, E, I correct
and correct durations

A1



A starting at 0 and ending at 3

M1

F starting at 6 and ending at 11
two correct with correct height

A1

G starting at 13 and ending at 14
all correct with correct height
withhold first A1 earned if it is not clear which activities
take place at any given time
withhold another A1 if "holes" appear in histogram

A1

5

(e) New earliest *J* 22 days
assuming activities continuous

B1

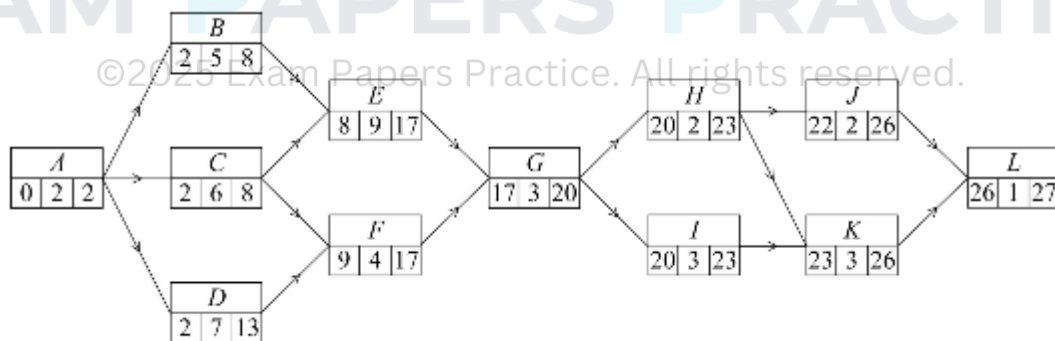
Minimum extra time 5 days
assuming activities continuous

B1

2

[14]

Q3.



(a) Forward pass
up to one slip ft

M1

Correct

A1

Backward pass

up to one slip ft

M1

Correct

A1

4

(b) (i) Critical path $A C E G I K L$

B1

(ii) Float for $D = 13 - 2 - 7$
'their 13' - 'their 2' - 7

M1

= 4 days

A1

3

(c) $A C E G I K L$ correct durations
 and heights }
one slip in duration or height

M1

correct

A1

D and B and F correct (no "holes")

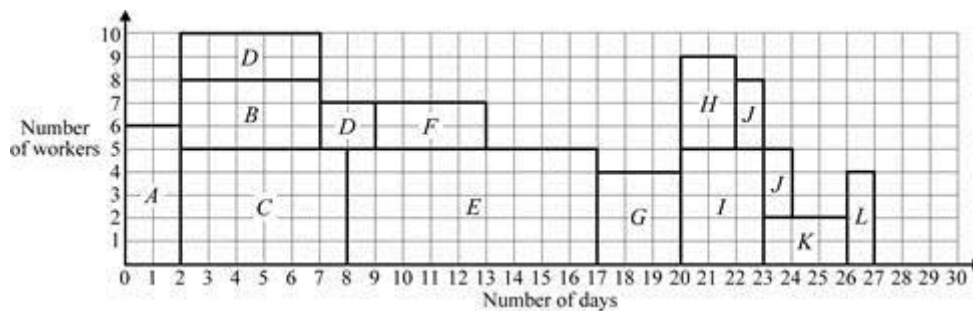
B1

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H and J correct (no "holes")

*withhold final mark earned if not clear which
 activities are taking place at any time*

B1



4

(d) Correctly dealing with D , B and F

ft 1 slip

B1

Correctly dealing with *H* and *J*

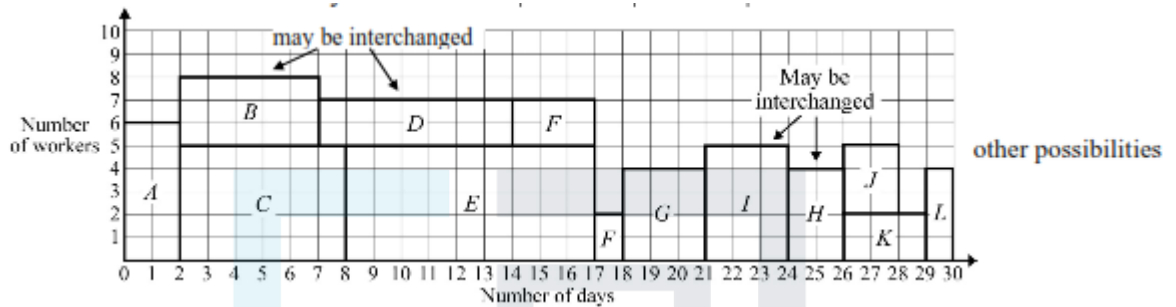
ft 1 slip

B1

Minimum extra time = 3 days

CAO

B1

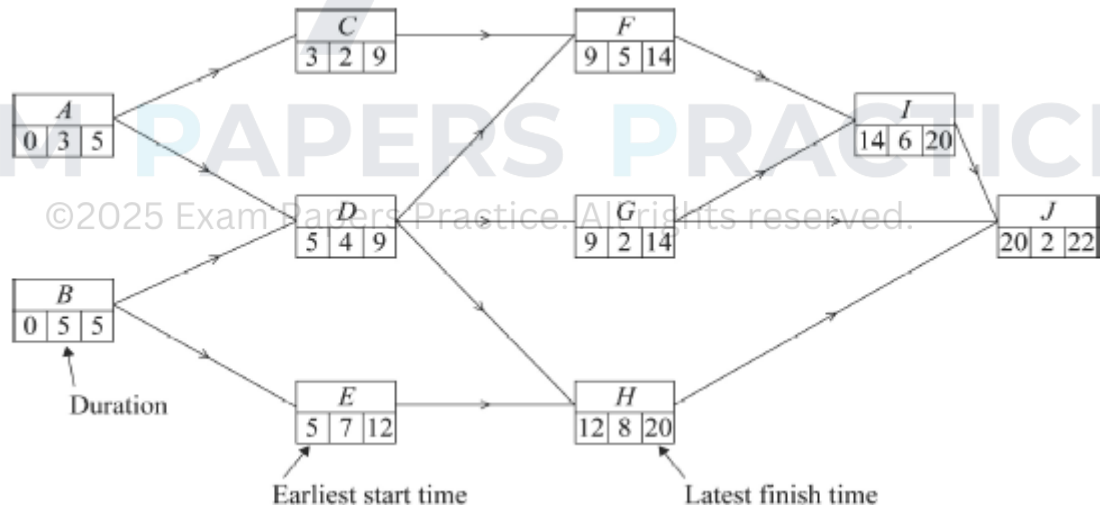


3

[14]

Q4.

(a)



(i) Earliest start times

Condone one slip with FT

M1

All correct

A1

2

(ii) Latest finish times

M1

All correct

A1

2

- (b) Critical paths *BEHJ*
BDFIJ
And no others

B1B1

Minimum completion time 22 days

B1

3

- (c) (i) *BEHJ* or *BDFIJ* correctly fitted
Withhold first B1 mark if activities not clearly indicated

B1

Second of critical paths

B1

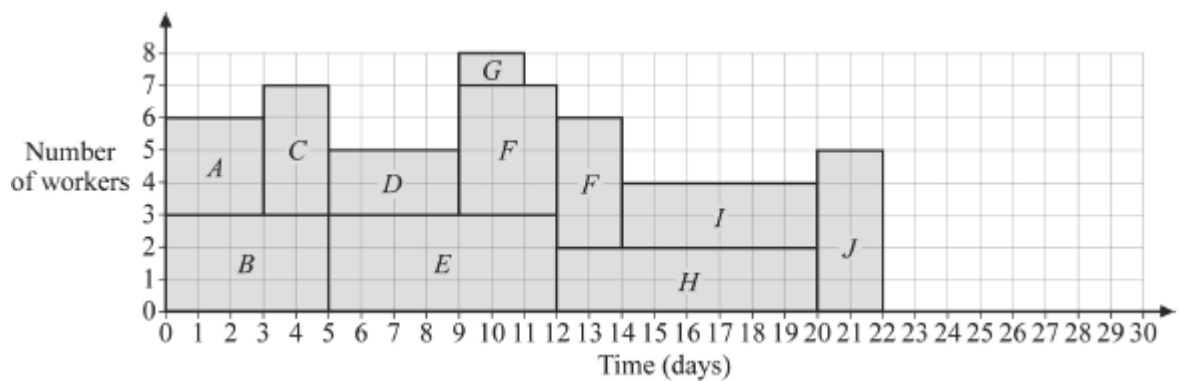
A and C shown correctly

B1

F and G shown correctly
no gaps in blocks etc

B1

4



- (ii) Problem with C - now starts day 5
E delayed until day 7
Or C starts day 12

E1

E delayed until day 7
F cannot start until day 14
 and *F* starts day 14

E1

(*H*, *I* and *J* delayed)
 Extra time required 5 days

B1

3

[14]

Q5.

- (a) *G*, *H* and *I* in correct place

M1

Lines (with arrows) correct

A1

2

- (b) Forward pass (no more than 1 error FT)

M1

Early start times correct

A1

Backward pass (no more than 1 error FT)

M1

Latest finish times correct

A1

See below

4

- (c) Correct critical path: *ACEGI*

B1

Correct minimum time: 25 days

B1

2

- (d) "Their" critical activities
See below

B1ft

Block $0 \leq t \leq 10$

B1

$10 \leq t \leq 11$

B1

All correct including labels
CSO

B1

4

(e) Problem with F or day 11

M1

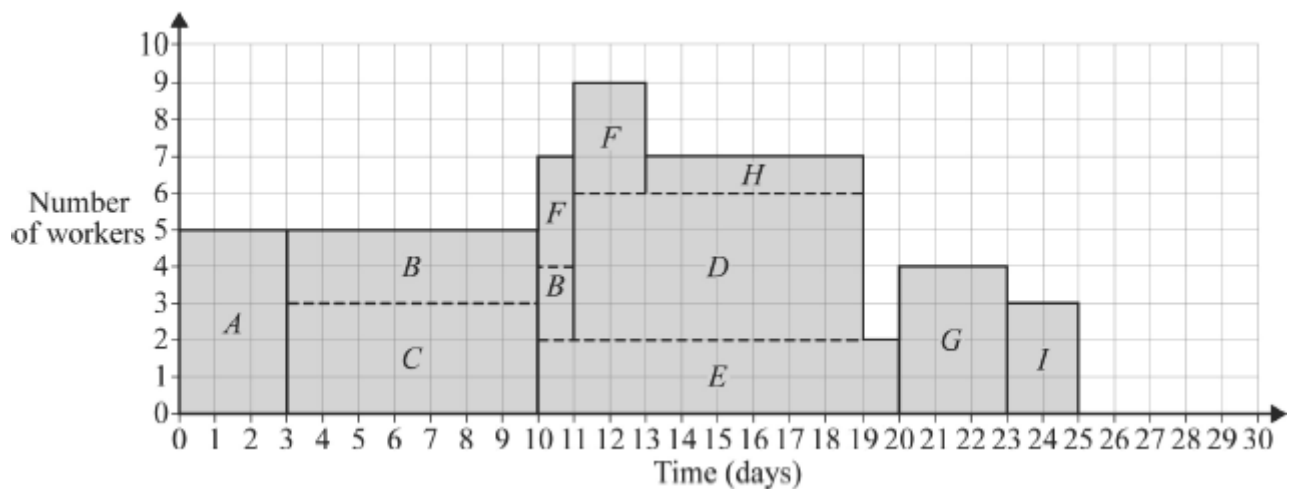
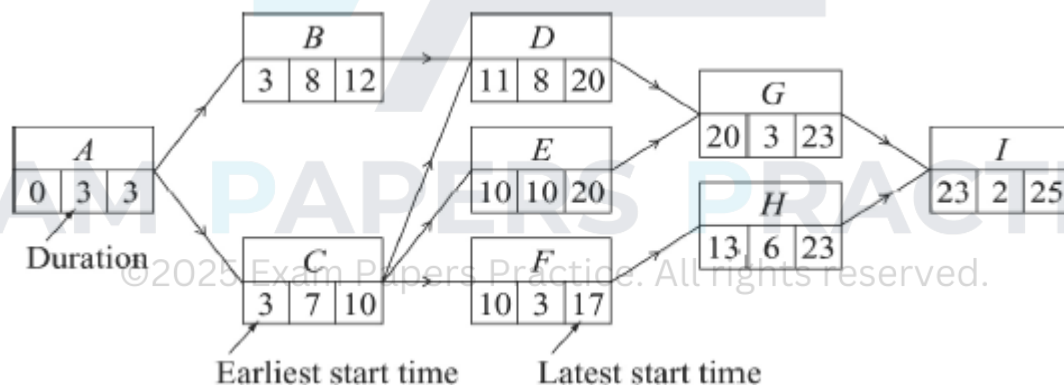
Delay start of D (by 2 days), then G and I (by 1 day)

A1

Extra time 1 day

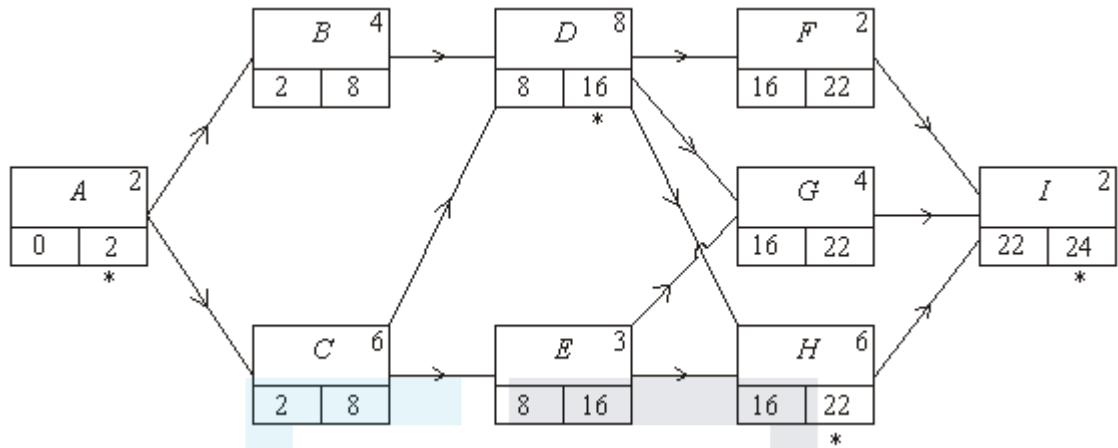
B1

3



Q6.

(a)



Activity network SCA

almost correct (up to 2 slips)

all correct

M1

A1

A1

3

(b) Forward pass for earliest times

M1

A1

2

(c) Backward pass

M1

A1

2

(d) Critical path is ACDHI

B1

Minimum completion 24 days

B1

2

(e) Non-critical

B E F G

Float

2 5 4 2

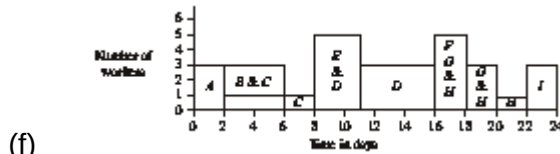
At least 3 activities and float in one activity ft correct

M1

ft their earliest and latest times

A1ft

2



Histogram ≤ 11

M1

Correct

A1

Rest as histogram – generally start activities ok

M1

All correct

A1

4

Resource histogram

(g) Problems with D & E solved by E coming after D

M1

Problem at 16-18 days with F can be solved by moving F to 20-22

A1

Must overrun by equivalent to duration of E (3 days)

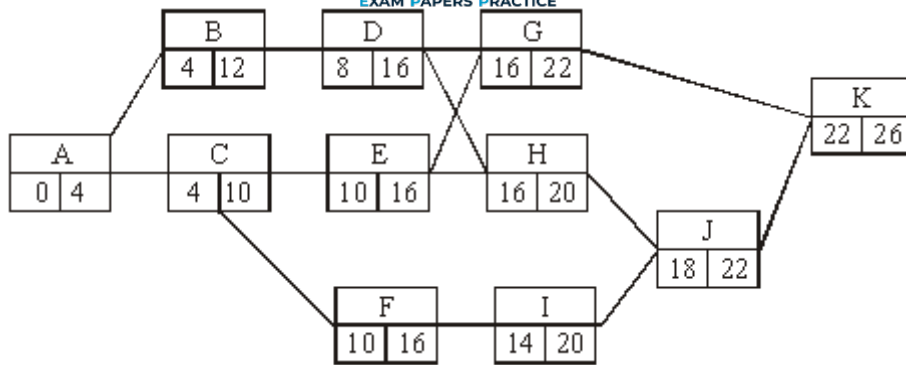
B1

3

[18]

Q7.

(a)



(i) Forward

M1A1

2

(ii) Back

M1A1

2

(b) CP = A C E G K
Min time = 26
Both

B1

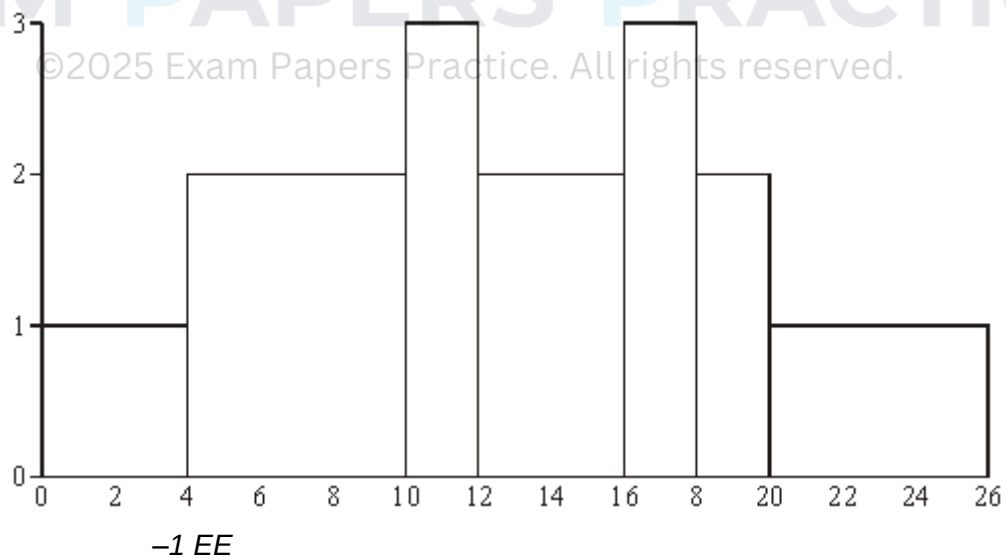
1

(c) B, D

B1

1

(d)



M1A2

3

(e) (i) Min time = 28

B1

1 worker critical path

other activities $\Rightarrow$ J takes until 24

$\therefore$ overrun = 2

E1

2

(ii) A C E G K

and

B D F (H) I J

H either

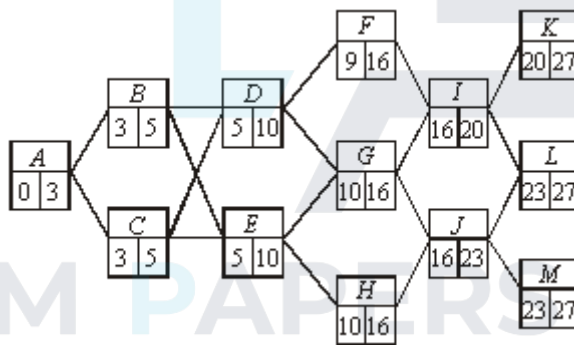
B1

1

[12]

Q8.

(a) (i) & (ii)



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Forward

M1

A1

2

Back

M1

A1

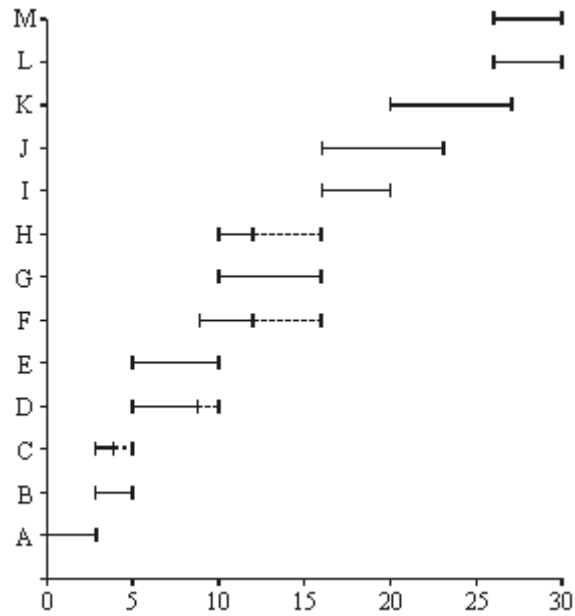
2

(b) C, H, F, D

B1

1

(c)



- (d) (i) Extra at KLM M1
- Min extra 4
Total = 31
- A1 3
- A2
- M1
- A1 2
- (ii) A B D F H I K
- C E G J L M
- B1
- OE
- B1 2
- [12]

Q9.

(a)

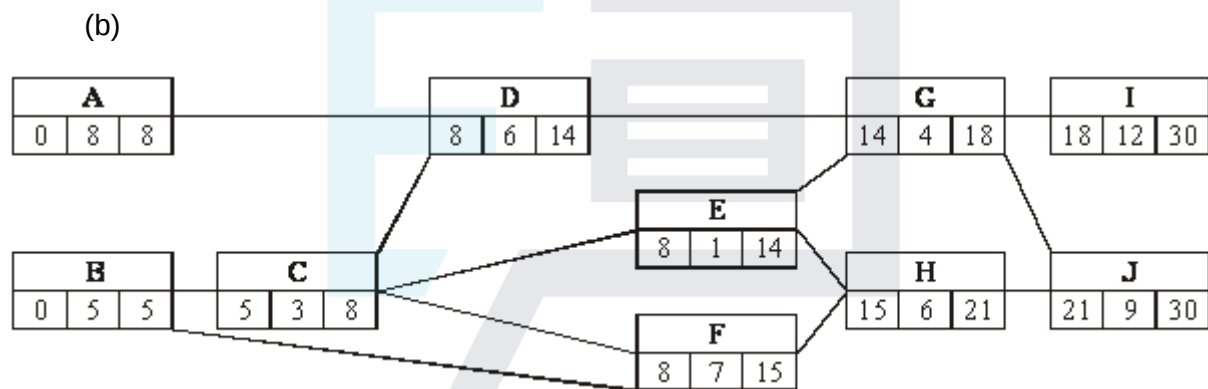
Activity	Immediate Predecessor
A	-
B	B

C	A, C
D	C
E	(B,) C
F	D, E
G	E, F
H	G
I	G, H
J	

-1 EE

B2, 1

2



Forward pass

Back pass

M1 A1

M1 A1

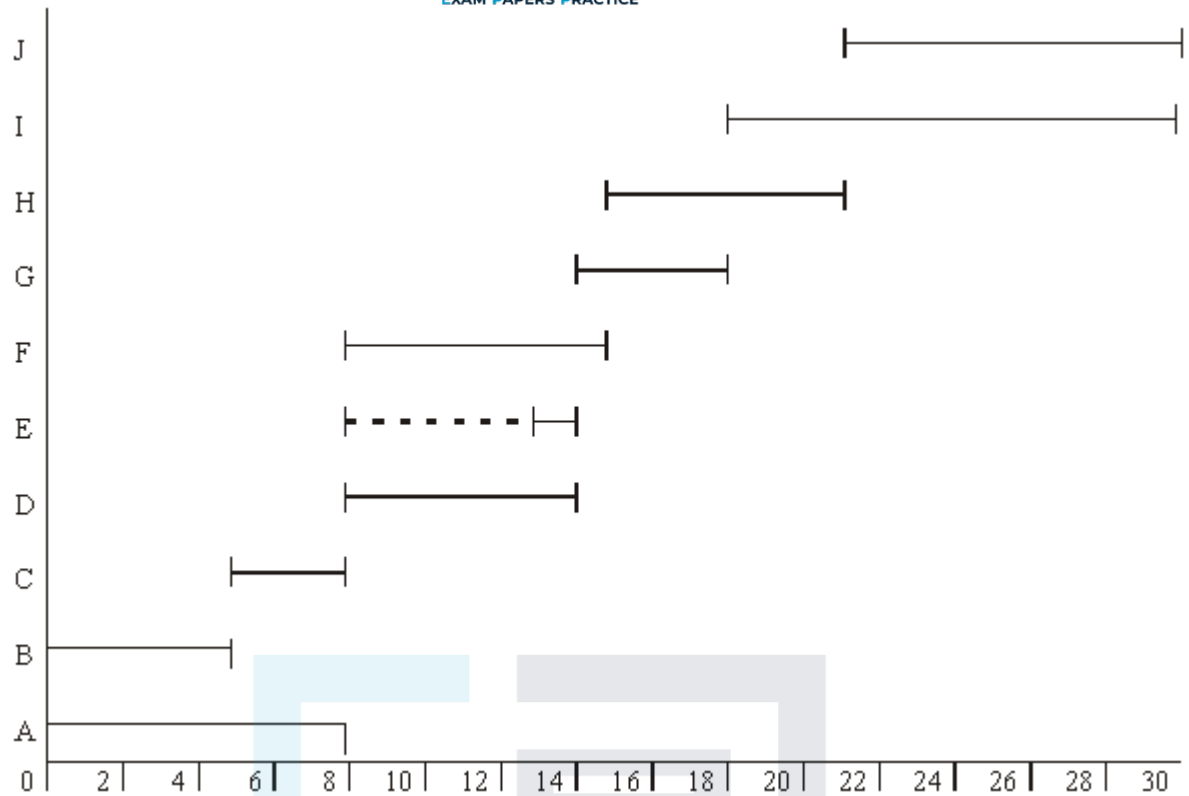
4

(c) Critical A, B, C, D, F, G, H, I, J

B1

1

(d)



-1 EE

M1 A3F

4

(e) Problem with D E F

M1

∴ overrun = 1 (hour)

A1

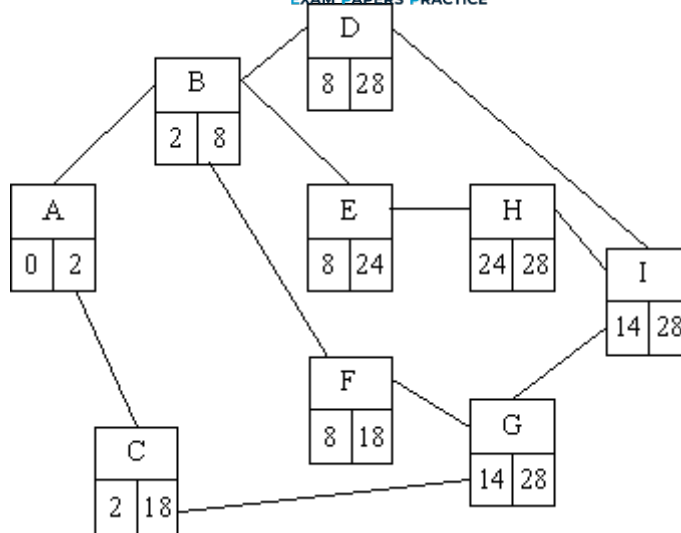
2

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[13]

Q10.

(a) (i)



-1 each independent error

M1 A3

4

(ii) As diagram

-1 EE

M1

A2

3

(iii) As diagram

-1 EE

M1

A2

3

(iv) A B E H I

Allow answers on diagram

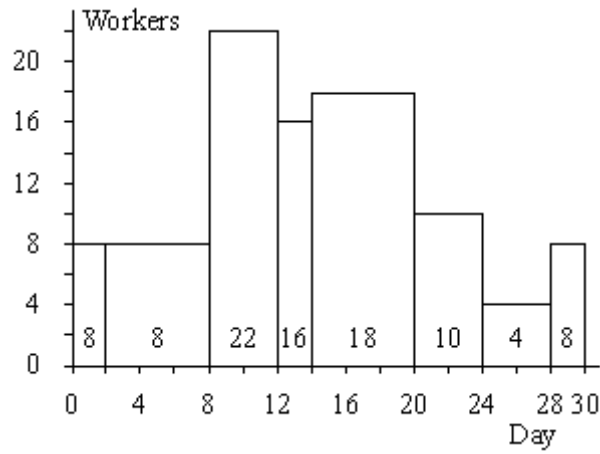
B1

30 days

B1

2

(b)



SCA

-1 EE

M1

A3

4

(c) Must reduce 12 → 8 by 4 workers

B1

4

Can't move C, E not enough float

B1

Can move D back 4 days to 8
or F and G back 4 days
6 days 4 days

B1

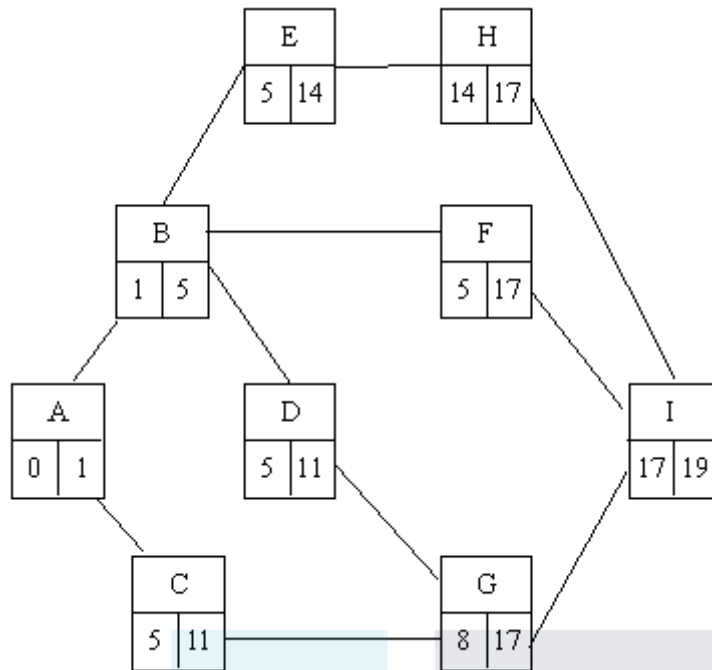
B1

EXAM PAPERS PRACTICE [20]

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Q11.

(a)



– 1 each independent error

(b) As diagram

M1A3

4

(c) As diagram

M1A2

3

(d) ABEHI

M1A2

3

19 days

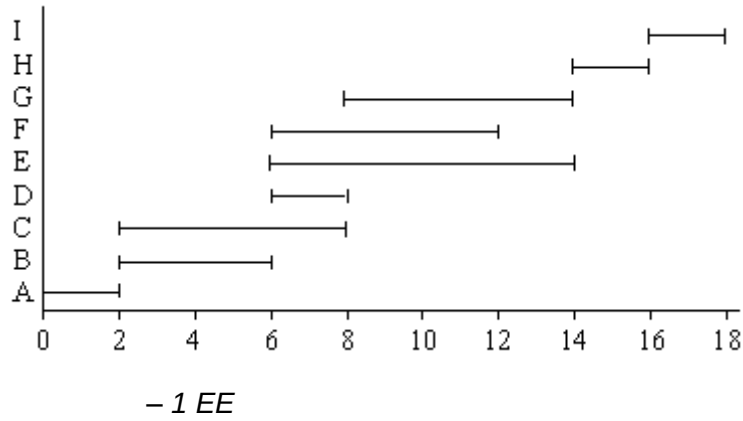
allow marked on diagram

B1

2

B1

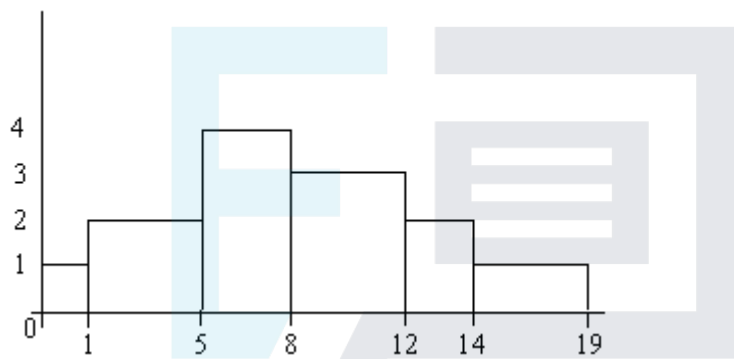
(e)



M1A3F

4

(f)



M1A1

2

(g) Worker 1 ABEHI

M1

2

Worker 2 CG

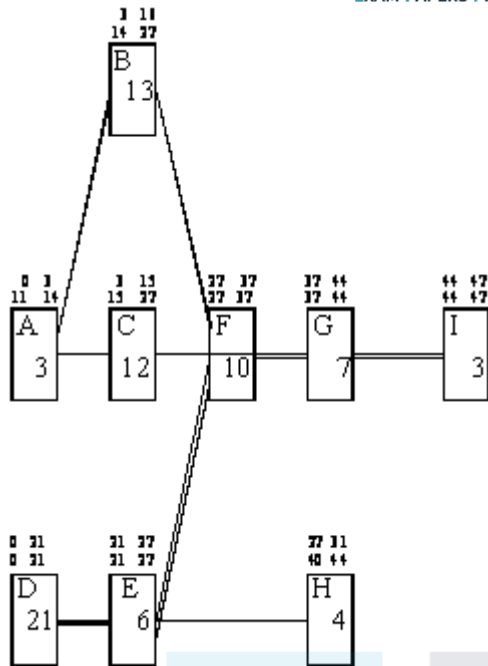
A1

Worker 3 DF

[20]

Q12.

(a)



Attempt at a network with all 9 activities included,
starting at A and D, finishing at I

All correct, – 1 EE

M1

A3

4

- (b) Upper figures show earliest times
Evidenced by labels

M1

(forward scan)

All correct

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A1

Lower figures show latest times
Evidenced by labels

M1

(backward scan)

All correct

A1

Critical path D-E-F-G-I

Critical path stated and shown. It c's labels provided upper
and lower figures match

A1FA1F

6

(c)

A	B	C	D	E	F	G	H	I
11	11	12	0	0	0	0	13	0
– 1 EE								

B2

2

(d) Cascade diagram

*Attempt a cascade diagram with all activities
and a suitable time scale*

M1

All correct. – 1 EE

A3

Minimum number of helpers: 3

No. of helpers (can be earned independently).

B1

5

(e) Least delay: 1 day

Delay

B1

Achieved by starting A and B as early as possible
then do C immediately after B

OE involving B and C sequential rather than parallel

E1

2

Examiner reports

Q1.

This question proved to be a good source of marks for all students, with many scoring full marks. There were problems in part (e) where some students left 'holes' with their blocks and others had the correct histogram profile, but failed to indicate the activities linked to each block.

Q2.

Part (a) Almost every candidate found the correct values of the constants x , y and z .

Part (b) The two critical paths were found correctly by most candidates, but several candidates wrote down extra paths that were not critical paths.

Part (c) Most candidates identified the correct activity with the largest float and calculated the float correctly.

Part (d) The histogram was usually constructed correctly but a few candidates left "holes" in their blocks and others made slips having certain blocks with an incorrect height or width.

It was necessary to indicate clearly which activities were taking place at any given time; those who had the correct histogram profile, but without the activities indicated, did not score full marks.

Part (e) The resource levelling caused difficulties for many, with only the best candidates finding the correct minimum extra time required and the new start time for activity J .

Q3.

In part (a) almost every candidate calculated the earliest start times and latest finish times correctly.

In part (b)(i) the single critical path was again found correctly by almost everyone.

In (b)(ii) a few candidates did not know how to calculate the float for activity D , but the majority were able to do so by finding $13 - 2 - 7$.

In part (c) the histogram was usually constructed correctly but a few candidates left gaps or "cantilevers" in their blocks and others made slips having certain blocks with an incorrect height or width. It was pleasing to see most candidates indicating clearly which activities were taking place at any given time.

In (d) the resource levelling caused a few problems; some could not deal correctly with the activity F which needed to be fitted in before G started. A number of alternatives were possible with the events following G , but the common mistake was trying to start K before H had finished. Those with a correct second histogram had no trouble in stating the correct minimum extra time required.

Q4.

In part (a) almost all the candidates calculated the earliest start times and latest finish times correctly.

In part (b) a number of even very able candidates included an extra incorrect critical path, namely *BDHJ*.

The histogram in part (c)(i) was not always constructed correctly, with quite a few candidates leaving gaps in their blocks. It was pleasing to see most candidates indicating clearly which activities were taking place at any given time. Explanations in part (c) (ii) were often muddled when trying to explain which activities would overrun. A common mistake was to suggest that activities such as G needed to be delayed when they did not. The best solutions indicated the new starting times for all the activities that needed to be delayed. Very few could justify that the new extra time required was 5 days.

Q5.

The network diagram was usually correct and almost all candidates calculated the earliest start times and latest finish times correctly and hence found the minimum completion time and critical path. Many candidates did not produce a correct resource histogram. The section between 10 and 12 days caused the biggest problem with many diagrams not having complete vertical blocks. It was good to see most candidates heeding the advice of previous reports in labelling their diagrams so that it was possible to see which activities were taking place on a particular day.

Q6.

Parts (a),(b),(c),(d) and (e) were very well done by practically all the candidates. The topic of critical path analysis seems to be well understood by most candidates. Problems started to creep in when answering part (f) where sometimes the resource histogram was not a histogram at all; being more like a random collection of rectangles piled on top of each other, rather like Stonehenge. Candidates had been expected to indicate which activities were being represented on the histogram so as to make resource levelling easier to identify. Many candidates were successful in finding the minimum extra time, although the explanations as to why three extra days are sufficient were often incomplete.

Q7.

This question was well answered by the majority of candidates. Marks were lost in part (d), where some candidates drew a Gantt chart rather than a resource histogram. Nearly all candidates used graph paper for their attempt at the resource histogram. Marks were lost in part (e)(ii) by candidates not stating clearly which worker should be allocated which task. The mark could not be obtained by quoting the critical path found in part (b), as the critical part (b) to part (e)(i).

Q8.

This question proved to be a good source of marks for all candidates, although a significant number of candidates failed to score full marks in part (c), due to the inaccuracies of their diagram or through drawing a resource histogram instead of a Gantt diagram.

Q9.

This question was well answered, with the majority of candidates scoring at least 10 out of 13 marks. Whereas parts (a), (b) and (c) were nearly always correct, candidates did make mistakes in part (d). The question stated that activities were to start as **late** as possible. This was the first time that this had been asked. A number of candidates made a mistake with activity *E*.

Also, for the first time, some candidates failed to show accurately the start and finish times

for their Gantt diagram. If candidates draw their answer in their answer booklet, they must clearly indicate start and finish times for each activity. Candidates who used graph paper did not lose accuracy marks.

Q10.

This was a standard critical path analysis question in which candidates were expected to use their diagram in part (a) to help them with the later parts of the question. It seems that candidates have been made aware of previous comments in examiners' reports and used their network as a working diagram. Part (c) proved to be a good discriminator with only the best candidates being able to justify their answers fully.

Q11.

The question was a routine application of critical path analysis for which the candidates were well prepared and scored nearly all available marks in the first four parts. It was pleasing to note that candidates worked their critical path on their network diagram with the benefit of saving time for the later, more thoughtful parts of the question. Part (g) proved to be too challenging for the candidates.

Q12.

The question was a routine application of critical path analysis for which the candidate was well prepared and scored nearly all the available marks.